CDKN2B fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDKN2B gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 2/3 fusions with known frame status (66.7%) were in-frame. 2/10 fusions (20.0%) retained the Ankyrin repeats (3 copies) domain. Domain retention was tested with Fisher's exact test (p=0.0222222, statistically significant at alpha=0.05). No literature benchmark is configured for CDKN2B.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 7 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 6 events.

Domain retention

Ankyrin repeats (3 copies) domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, p=0.0222222 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.207921.

Domain disruption

Domain-disruption analysis was skipped: CDKN2B has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

57% of 7 fusions with this gene as the 5' partner join at exon 2, against 52% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=1, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 53 aa (exon 2; label-permutation p=0.743257). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -1 aa from the nearest configured domain boundary at 54 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=2) and "not_retained" (n=8). For Frame_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not_retained: 0/1

(0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 11.5 vs 48.25, p=0.170234 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for CDKN2B in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (3 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: CDKN2B has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: CDKN2B has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (20 row(s) in window_scan, 8 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CDKN2A	6	0.5	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.2992292754929425	1
CDKN2B-AS1	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	2
CNTRL	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	3
ELAVL2	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	4
GAB2	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	5
MTAP	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	6
SLC24A2	1	0.08333333333333333	0.06821020790487234	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.10692158318525016	7

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
38	0	2	1	7	0.0	1.0	-0.0
52	0	2	3	5	0.0	1.0	-0.0
53	2	0	3	5	None	0.44444444444444453	0.3521825181113624
69	2	0	4	4	None	0.4666666666666668	0.3309932190414243
73	2	0	5	3	None	1.0	-0.0
75	2	0	6	2	None	1.0	-0.0
113	2	0	7	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

2	0
0	8

frame_domain_contingency_tables_by_tumor_type

stratum	table
CCRCC	[0, 0], [0, 1]
GIST	[0, 0], [0, 2]
HCC	[0, 0], [0, 1]
LUAD	[1, 0], [0, 1]
OS	[0, 0], [0, 1]
PAAD	[0, 0], [0, 1]
PRAD	[1, 0], [0, 0]
UMEC	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Ankyrin repeats (3 copies)	10	2	5	3	0.34104938271604934

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	2	12	0.16666666666666666

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CDKN2A	6	6	6	6
CDKN2B-AS1	1	1	1	1
CNTRL	1	1	1	1
ELAVL2	1	1	1	1
GAB2	1	1	1	1
MTAP	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
SLC24A2	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
9	22006000	22007000	5	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
9	22008000	22009000	4	EVT-P-0002886-T01-IM3-40, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
9	22007000	22008000	1	EVT-P-0038263-T01-IM6-11

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
52	2	2	489	False
53	2	2	42	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	2	4	7	0.5714285714285714	0.5177217149048134	0.2686388450457373
five_prime	1	3	7	0.42857142857142855	0.4822782850951865	0.12880911804780243
three_prime	2	3	3	1.0	0.9123974616932363	0.11944779482718283

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-147	53	200	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
-131	69	200	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
-47	53	100	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
-31	69	100	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
3	53	50	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
19	69	50	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
28	53	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
38	63	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
44	69	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0060581-T01-IM7-13
48	73	25	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
53	78	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
53	103	50	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
63	113	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
64	114	50	5	5	0	2	0.0	0.44444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
69	119	50	5	5	0	2	0.0	0.44444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
69	169	100	5	5	0	2	0.0	0.44444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
69	269	200	5	5	0	2	0.0	0.44444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
73	123	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
73	173	100	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
73	273	200	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
28	53	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, EVT-P-0060581-T01-IM7-13
44	69	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0060581-T01-IM7-13
53	78	25	5	5	2	0	None	0.44444444444444453	0.3521825181113624	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38
64	114	50	5	5	0	2	0.0	0.44444444444444453	0.3521825181113624	EVT-P-0009347-T01-IM5-34, EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
48	73	25	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0053040-T01-IM6-38, ... (6 total)
19	69	50	6	4	2	0	None	0.4666666666666668	0.3309932190414243	EVT-P-0002886-T01-IM3-40, EVT-P-0010183-T02-IM6-12, EVT-P-0038263-T01-IM6-11, EVT-P-0042985-T01-IM6-37, EVT-P-0045616-T01-IM6-36, ... (6 total)
63	113	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0010183-T02-IM6-12, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38
73	123	50	4	6	0	2	0.0	0.4666666666666668	0.3309932190414243	EVT-P-0009347-T01-IM5-34, EVT-P-0035576-T01-IM6-1, EVT-P-0045019-T02-IM6-10, EVT-P-0053040-T01-IM6-38

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003557 6-T01-IM6-1	P-003557 6-T01-IM6	P-003557 6	Gastrointestinal Stromal Tumor	GIST	CDKN2A-CDKN2B	CDKN2A	unknown	five_prime	22006179	2	75	False	74	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.2716049382716049	True	partner_lacks_domain	PF12796					66			True				Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.224:CDKN2B_c.153:CDKN2Adel | ProteinFusion: mid-exon {CDKN2B:CDKN2A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004501 9-T02-IM6-10	P-004501 9-T02-IM6	P-004501 9	Non-Small Cell Lung Cancer	LUAD	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22006065	2	113	False	112	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.740740740

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crossscheck_protein_position	fusion_annotation_crossscheck_agrees	fusion_annotation_crossscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003826 3-T01-IM6-11	P-003826 3-T01-IM6	P-003826 3	Endometrial Cancer	UMEC	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22007679	1	52	True	52	True	exact match	lost	0.0	False													Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A(NM_000077) Rearrangement: c.156+1118:CDKN2B_c.192:CDKN2Adel | Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001013-T02-IM6-12	P-001013-T02-IM6	P-001013	Gastrointestinal Stromal Tumor	GIST	CDKN2B-CDKN2A	CDKN2A	unknown	five_prime	22006197	2	69	False	68	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.19753086419753085	True	partner_lacks_domain	PF12796					65			False				Ankyrin repeats (3 copies)	CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.206:CDKN2B_c.194-11:CDKN2Adel | Protein Fusion: mid-exon {CDKN2B:CDKN2A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006058 1-T01-IM7-13	P-006058 1-T01-IM7	P-006058 1	Hepatobiliary Cancer	HCC	CDKN2B-CDKN2A	CDKN2A	out-of-frame	five_prime	22008168	1	52	True	52	True	exact match	lost	0.0	False													Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - CDKN2A (NM_000077) rearrangement: c.156+629:CDKN2B_c.458-143:CDKN2Adel | Protein Fusion: out of frame {CDKN2B:CDKN2A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000934 7-T01-IM5-34	P-000934 7	P-000934 7	Renal Cell Carcinoma	CCRCC	CDKN2B-CDKN2B-AS1	CDKN2B-AS1	unknown	five_prime	220060 62	2	114	False	113	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.753086419753086 4	True	partner_unmapped	PF12796									CDKN2B-AS1 fusion has ambiguous genomic breakpoints within target locus 21994790-22121094: Site1=22006062, Site2=22015263			Ankyrin repeats (3 copies)	CDKN2B (NM_004936) rearrangement: c.341_chr9:g.22015263inv | Protein fusion: mid-exon (CDKN2B-CDKN2BAS)	NA

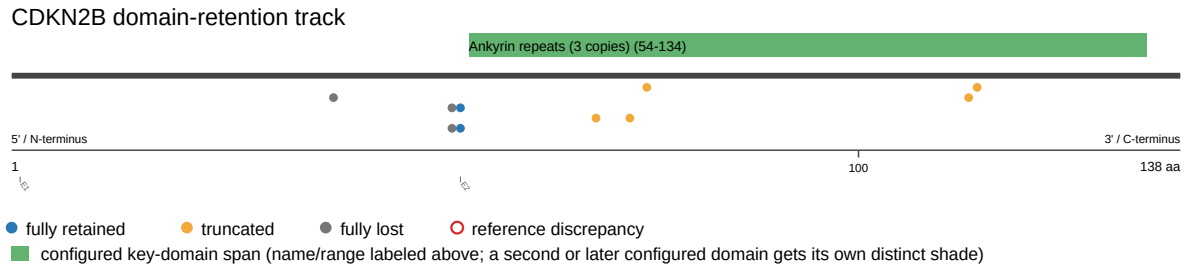
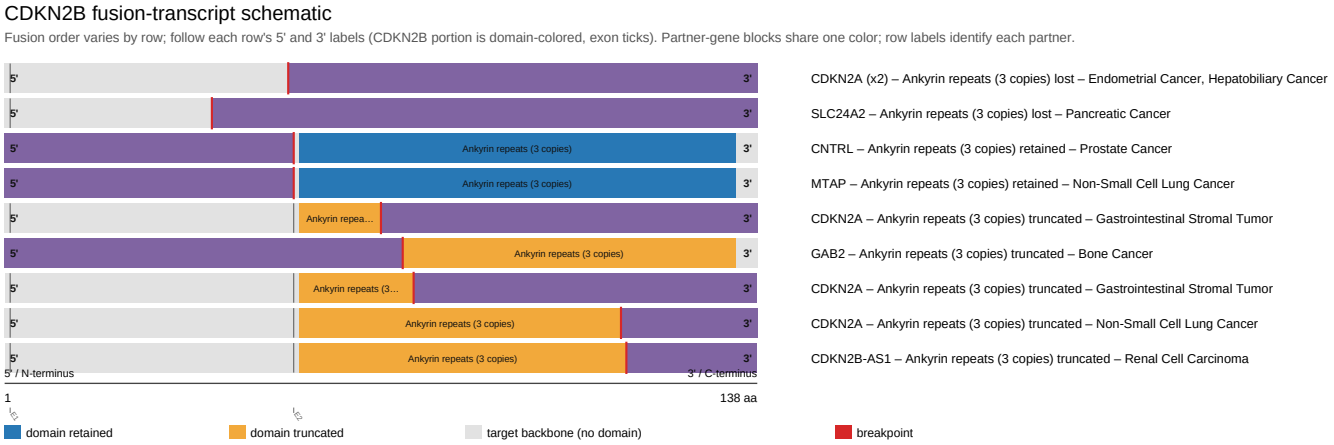
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004561 6-T01-IM6-36	P-004561 6-T01-IM6	P-004561 6	Pancreatic Cancer	PAAD	CDKN2B-SLC24A2	SLC24A2	unknown	five_prime	220088 40	1	38	False	37	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False															Ankyrin repeats (3 copies)		CDKN2B (NM_004936) - SLC24A2 (NM_020344) rearrangement: c.113:CDKN2B_c.931-51322:SLC24A2del | Protein Fusion: mid-exon { CDKN2B:SLC24A2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004285-T01-IM6-37	P-004285-T01-IM6	P-004285	Prostate Cancer	PRAD	CNTRL-CDKN2B	CNTRL	in-frame	three_prime	22008250	2	53	True	53	True	exact match	retained	1.0	False													Ankyrin repeats (3 copies)				CNTRL (NM_007018) - CDKN2B (NM_004936) rearrangement: c.1455+1114:CNTRL_c.156+547:CDKN2Binv | Protein Fusion: in frame {CNTRL:CDKN2B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005304-T01-IM6-38	P-005304-T01-IM6	P-005304	Bone Cancer	OS	GAB2-CDKN2B	GAB2	unknown	three_prime	220061844	2	73	False	73	True	exact match	disrupted	0.765432098765432 1	True	partner_lacks_domain	PF12796								False				Ankyrin repeats (3 copies)	GAB2 (NM_080491) - CDKN2B (NM_004936) rearrangement: t(9;11)(p21.3;q14.1)(chr9:g.22006184::chr11:g.78068634)| Protein Fusion: mid-exon { GAB2:CDKN2B}	NA
EVT-P-002886-T01-IM3-40	P-002886-T01-IM3	P-002886	Non-Small Cell Lung Cancer	LUAD	MTAP-CDKN2B	MTAP	in-frame	three_prime	22008292	2	53	True	53	True	exact match	retained	1.0	False												Ankyrin repeats (3 copies)			NA | Protein fusion: in frame (MTAP-CDKN2B)	NA

Figures

domain retention outliers



fusion schematic